Adaptive Learning Rates: Momentum, RMSProp and Adam

Plain stochastic gradient descent uses one learning rate for every parameter. Adaptive methods scale each parameter's step by statistics of its past gradients, which helps when gradients are sparse or badly scaled.

1. Momentum
A velocity term accumulates past gradients, so updates keep moving through flat regions and oscillations across steep valleys cancel out.

2. RMSProp
Each step is divided by a running average of squared gradients.

3. Adam
Adam keeps both a running mean of the gradients (first moment) and of their squares (second moment), corrects both for their bias towards zero early in training, and steps by their ratio.

4. Practical defaults

A learning rate of 0.001 with beta1 = 0.9, beta2 = 0.999 and epsilon = 1e-8 works for most problems. Warm-up and decay schedules still matter for large models.